

Subgroups and Homomorphisms

Week 2 · Lecture 3

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Group Theory

Reading for Lecture 3

- **Zee, I.1, p. 43 and pp. 49–50:** cyclic subgroups, homomorphisms and isomorphisms.
- **Zee, I.2, p. 55:** permutations and Cayley's theorem.
- **Bünemann, §2.1 and §2.3.4:** element order, isomorphisms and concrete subgroups.

Kernel and image are developed here through definitions and examples. Zee returns to the kernel in I.2, p. 64. We use Z_n for cyclic groups and e for the identity. Products of transformations act from right to left.

Group Theory

Groups inside groups

A subset $H \subseteq G$ is a **subgroup**, written $H \leq G$, if the same operation makes H a group.

$$\{e, r, r^2\} \leq D_3, \quad \{e, s\} \leq D_3.$$

For a subgroup, the identity and inverses are inherited from G .

Group Theory

The subgroup test

A nonempty subset $H \subseteq G$ is a subgroup if it is closed under products and inverses:

$$a, b \in H \implies ab \in H, \quad a \in H \implies a^{-1} \in H.$$

Associativity comes from G . For any $a \in H$, closure under inverses gives $a^{-1} \in H$ and then $aa^{-1} = e \in H$.

Group Theory

An equivalent subgroup test

A nonempty subset $H \subseteq G$ is a subgroup exactly when

$$a, b \in H \implies ab^{-1} \in H.$$

To see why this single condition suffices:

- Set $b = a$ to obtain $e \in H$.
- Set $a = e$ to obtain $b^{-1} \in H$.
- Apply the condition to a and b^{-1} to obtain $ab \in H$.

Conversely, every subgroup satisfies the condition by closure under inverses and products.

Group Theory

Example: subgroups of Z_6

The operation is addition modulo 6.

$$H_2 = \{0, 2, 4\}, \quad H_3 = \{0, 3\}.$$

Both contain 0 and are closed under addition and additive inverses modulo 6. But $\{0, 2\}$ fails closure because $2 + 2 = 4 \notin \{0, 2\}$.

Group Theory

Generated subgroups

The subgroup generated by g consists of all integral powers of g :

$$\langle g \rangle = \{g^k : k \in \mathbb{Z}\}.$$

For finite order m , this becomes $\{e, g, \dots, g^{m-1}\}$, so $|\langle g \rangle| = \text{ord}(g) = m$. In additive notation, $\langle 2 \rangle \leq Z_6$ means $\{0, 2, 4\}$.

Group Theory

All subgroups of Z_6

$$\{0\}, \quad \{0, 3\} = \langle 3 \rangle, \quad \{0, 2, 4\} = \langle 2 \rangle, \quad Z_6 = \langle 1 \rangle.$$

Their orders are 1, 2, 3, 6. Why is this list complete? A subgroup containing 1 or 5 is all of Z_6 . One containing 2 or 4 contains $\{0, 2, 4\}$; if it also contains 3, it contains $3 - 2 = 1$ and is all of Z_6 . Different generators can give the same subgroup:

$$\langle 4 \rangle = \langle 2 \rangle, \quad \langle 5 \rangle = \langle 1 \rangle.$$

Group Theory

Subgroups of the triangle group

In $S_3 \cong D_3$ we have

$$\{e\}, \quad \langle(123)\rangle = \{e, (123), (132)\},$$

$$\langle(12)\rangle, \quad \langle(13)\rangle, \quad \langle(23)\rangle, \quad S_3.$$

Each transposition generates a different subgroup of order 2. The set $\{e, (123), (23)\}$ is not a subgroup: $(123)(23) = (12)$ lies outside it.

Textbook: Compare Bünemann, §2.3.4, p. 21, where r is δ_3 and the three order-two elements are δ_{2i} .

Group Theory

A useful closure property

If H and K are subgroups of G , then $H \cap K$ is a subgroup. It contains e ; products and inverses stay in both H and K .

$$\langle (12) \rangle \cap \langle (23) \rangle = \{e\}.$$

The union $H \cup K$ need not be a subgroup. Here $(12), (23) \in H \cup K$, but $(12)(23) = (123) \notin H \cup K$.

Group Theory

Maps that preserve multiplication

A **homomorphism** $\varphi : G \rightarrow K$ obeys

$$\varphi(ab) = \varphi(a)\varphi(b) \quad (a, b \in G).$$

It respects the operations of both groups. In additive notation this reads $\varphi(a + b) = \varphi(a) + \varphi(b)$. A homomorphism may send several elements of G to the same element of K .

Group Theory

When is a homomorphism an isomorphism?

An **isomorphism** is a homomorphism that is both one-to-one and onto. It identifies two descriptions of the same abstract group. For $\omega = e^{2\pi i/3}$,

$$\mathbb{Z}_3 \longrightarrow \{1, \omega, \omega^2\}, \quad [k]_3 \longmapsto \omega^k.$$

This is well defined because $\omega^3 = 1$. It is bijective and preserves the operations:

$$\omega^{a+b} = \omega^a \omega^b.$$

Textbook: Zee, I.1, p. 50; Bünemann, §2.1, p. 12.

Group Theory

Identity and inverses under a homomorphism

Let e_G, e_K be the two identities. Since $\varphi(e_G)^2 = \varphi(e_G)$, multiplying by the inverse gives

$$\varphi(e_G) = e_K.$$

Apply φ to $aa^{-1} = e_G$:

$$\varphi(a^{-1}) = \varphi(a)^{-1}.$$

We did not need to put these rules into the definition: they follow from multiplication preservation.

Group Theory

Reducing residues modulo 3

Define $\varphi : Z_6 \rightarrow Z_3$ by $\varphi([a]_6) = [a]_3$.

$$\varphi([a]_6 + [b]_6) = [a + b]_3 = \varphi([a]_6) + \varphi([b]_6).$$

This is well defined because changing a by a multiple of 6 does not change its residue modulo 3.

$$0, 3 \mapsto 0; \quad 1, 4 \mapsto 1; \quad 2, 5 \mapsto 2.$$

Group Theory

Kernel and image

For a homomorphism $\varphi : G \rightarrow K$, define

$$\ker \varphi = \{g \in G : \varphi(g) = e_K\}, \quad \operatorname{im} \varphi = \{\varphi(g) : g \in G\}.$$

For $Z_6 \rightarrow Z_3$, the kernel is $\{0, 3\}$ and the image is all of Z_3 . The kernel records which distinctions in G the map forgets.

Group Theory

The kernel and image are subgroups

They are nonempty: $e_G \in \ker \varphi$ and $e_K = \varphi(e_G) \in \operatorname{im} \varphi$. If $a, b \in \ker \varphi$, then

$$\varphi(ab^{-1}) = e_K e_K^{-1} = e_K,$$

so $ab^{-1} \in \ker \varphi$. If $x = \varphi(a)$ and $y = \varphi(b)$ belong to the image, then

$$xy^{-1} = \varphi(ab^{-1}) \in \operatorname{im} \varphi.$$

The equivalent subgroup test therefore gives

$$\ker \varphi \leq G, \quad \operatorname{im} \varphi \leq K.$$

Group Theory

Injectivity and the kernel

$$\boxed{\varphi \text{ is injective} \iff \ker \varphi = \{e_G\}.}$$

If φ is injective and $\varphi(g) = e_K = \varphi(e_G)$, then $g = e_G$. Conversely, suppose $\ker \varphi = \{e_G\}$. Then

$$\begin{aligned}\varphi(a) = \varphi(b) &\implies \varphi(ab^{-1}) = e_K \\ &\implies ab^{-1} = e_G \\ &\implies a = b.\end{aligned}$$

Thus the map is injective.

Group Theory

Injective homomorphisms preserve element orders

Let $\varphi : G \rightarrow K$ be an injective homomorphism. For every positive integer m ,

$$\varphi(g)^m = e_K \iff \varphi(g^m) = \varphi(e_G) \iff g^m = e_G.$$

Hence the smallest such positive integer is the same:

$$\boxed{\text{ord}(\varphi(g)) = \text{ord}(g).}$$

If no such integer exists, both elements have infinite order. In particular, an isomorphism preserves element orders. This will distinguish the two groups of order four.

Group Theory

Rotations versus reflections

For the planar triangle group, define

$$\chi : D_3 \rightarrow \{+1, -1\}, \quad \chi(r) = +1, \quad \chi(s) = -1.$$

Rotations preserve orientation; reflections reverse it. Composition multiplies these signs, so χ is a homomorphism with kernel $\{e, r, r^2\} \cong Z_3$.

Treating each sign as a 1×1 matrix gives a **one-dimensional representation**.

In Bünnemann's prism realization, χ distinguishes threefold rotations from half-turns. All six spatial transformations have determinant $+1$; χ is not that determinant.

Group Theory

A tempting map that usually fails

Consider $f(g) = g^2$. If f were a homomorphism, we would have

$$(ab)^2 = a^2b^2 \quad \text{for every } a, b.$$

In D_3 take $a = r$, $b = s$: $(rs)^2 = e$ but $r^2s^2 = r^2 \neq e$. Squaring *is* a homomorphism on an Abelian group, where the factors commute.

Group Theory

Exercise: all homomorphisms $Z_6 \rightarrow Z_3$

Both groups use addition.

1. Explain why $\varphi([1]_6)$ determines the entire map.
2. Find all possible homomorphisms.
3. Check that each proposed map is well defined.
4. Find the kernel and image of each map.

Hint: write $[k]_6$ as a sum of copies of $[1]_6$. Your list should include the reduction map and $f([k]_6) = [2k]_3$.

Group Theory

Determining all three maps

Write $\varphi([1]_6) = [a]_3$. Then

$$\varphi([k]_6) = [ak]_3, \quad a = 0, 1, 2.$$

There are at most three possibilities. All three exist: replacing k by $k + 6m$ changes ak by $6am$, which vanishes modulo 3. Each map also preserves addition:

$$[a(k + \ell)]_3 = [ak]_3 + [a\ell]_3.$$

General rule: for a chosen $[a]_n \in Z_n$, the formula $[k]_m \mapsto [ak]_n$ defines a homomorphism exactly when $ma \equiv 0 \pmod{n}$. The condition follows from $[m]_m = [0]_m$ and ensures independence of the representative.

Group Theory

The three kernels and images

Map	Kernel	Image
$\varphi_0([k]_6) = [0]_3$	Z_6	$\{0\}$
$\varphi_1([k]_6) = [k]_3$	$\{0, 3\}$	Z_3
$\varphi_2([k]_6) = [2k]_3$	$\{0, 3\}$	Z_3

For φ_2 , the condition $2k \equiv 0 \pmod{3}$ is equivalent to $k \equiv 0 \pmod{3}$. The last two maps have the same kernel and image but are different: $\varphi_1([1]_6) = [1]_3$, whereas $\varphi_2([1]_6) = [2]_3$.

Group Theory

Cayley: multiplication produces permutations

For fixed $g \in G$, define $L_g : G \rightarrow G$ by $L_g(x) = gx$. This map is a bijection because

$$L_{g^{-1}}L_g = L_gL_{g^{-1}} = \text{id}_G .$$

Thus L_g is a permutation of the elements of G . For example, in $Z_3 = \{e, r, r^2\}$,

$$L_r : \quad e \mapsto r, \quad r \mapsto r^2, \quad r^2 \mapsto e.$$

Write $\text{Sym}(G)$ for the group of all permutations of the set G .

Textbook: Zee, I.2, p. 55.

Group Theory

Cayley's theorem

Define $\Phi : G \rightarrow \text{Sym}(G)$ by $\Phi(g) = L_g$. It is a homomorphism because, for every $x \in G$,

$$(L_g L_h)(x) = g(hx) = (gh)x = L_{gh}(x).$$

Its kernel is trivial:

$$L_g = \text{id}_G \implies g = L_g(e) = e.$$

Therefore Φ is injective and $G \cong \text{im } \Phi$.

$|G| = n \implies G \text{ is isomorphic to a subgroup of } S_n.$

Group Theory

The rearrangement theorem

Multiplying every element of a group by a fixed element **permutes the group elements**. For finite $G = \{g_1, \dots, g_n\}$ and fixed $a \in G$,

$$\{ag_1, \dots, ag_n\} = G,$$

with every element appearing once. The same holds for right multiplication $g \mapsto ga$, whose inverse is $g \mapsto ga^{-1}$. Consequently, for any scalar-valued function F on a finite group,

$$\sum_{g \in G} F(ag) = \sum_{g \in G} F(g) = \sum_{g \in G} F(ga).$$

We will use this to average over groups in representation theory.

Group Theory

From subgroups to counting

Our examples have subgroup sizes that divide the group size:

$$|S_3| = 6, \quad |\langle(123)\rangle| = 3, \quad |\langle(12)\rangle| = 2.$$

Next lecture, cosets will explain this divisibility for every subgroup of a finite group. We will also construct larger groups using direct products.

Reading: Zee, I.1, p. 43 and pp. 49–50; I.2, p. 55. Bünemann, §2.1 and §2.3.4.

Cosets and Lagrange's Theorem and Direct Products

Week 2 · Lecture 4

Mehmet Özkan

Group Theory

Reading for Lecture 4

- **Zee, I.1, pp. 43–44:** cyclic subgroups and Lagrange's counting argument.
- **Zee, I.1, pp. 44–45:** direct products and Klein's four-group.
- **Bünemann, §2.3.4, pp. 21–22:** left cosets, equal coset size and the index.

Zee uses right cosets Hg in his counting proof; we use left cosets gH , following Bünemann. For direct products of groups we write $G \times K$. Zee writes $G \otimes K$ in this section.

Group Theory

Translating a subgroup

For $H \leq G$ and $g \in G$, the **left coset** containing g is

$$gH = \{gh : h \in H\}.$$

For additive groups, write $g + H = \{g + h : h \in H\}$. Since $e \in H$, we have $g = ge \in gH$. Thus the cosets cover G . A coset need not itself be a subgroup. Its role is to collect group elements into equal-sized pieces.

Group Theory

An example in Z_6

Let $H = \{0, 3\} \leq Z_6$. The distinct cosets are

$$0 + H = \{0, 3\}, \quad 1 + H = \{1, 4\}, \quad 2 + H = \{2, 5\}.$$

For example, $3 + H = 0 + H$, and $4 + H = 1 + H$. They are disjoint, each has two elements, and together they exhaust Z_6 .

Group Theory

Translating H by one of its own elements

If $h \in H$, then $\boxed{hH = H}$. **First inclusion:** if $k \in H$, closure gives $hk \in H$. Hence $hH \subseteq H$. **Reverse inclusion:** every $h' \in H$ can be written as

$$h' = h(h^{-1}h').$$

Since $h^{-1}h' \in H$, this shows $h' \in hH$. Therefore $H \subseteq hH$. This identity supplies a step in the coset proof.

Group Theory

Every coset has $|H|$ elements

The map $H \rightarrow gH$ defined by $h \mapsto gh$ is surjective by construction. It is injective because

$$gh_1 = gh_2 \implies h_1 = h_2$$

after multiplying on the left by g^{-1} . Therefore $|gH| = |H|$ for finite H .

Group Theory

Why cosets do not overlap partially

Suppose g_1H and g_2H share some x . Write $x = g_1h_1 = g_2h_2$. Then

$$g_2^{-1}g_1 = h_2h_1^{-1} \in H.$$

Thus $g_1 = g_2h$ for some $h \in H$, and

$$g_1H = g_2hH = g_2H,$$

using $hH = H$. **Two left cosets are either identical or disjoint.** Together with $g \in gH$, this proves that distinct cosets partition G .

Group Theory

Lagrange's theorem

The distinct left cosets partition G . If there are $[G : H]$ cosets and G is finite, each has $|H|$ elements. Hence

$$|G| = [G : H] |H|.$$

The number $[G : H]$ is the **index** of H in G . In particular, $|H|$ divides $|G|$.

Textbook: Zee, I.1, pp. 43–44; Bünemann, §2.3.4, pp. 21–22.

Group Theory

Lagrange on S_3

Let $H = \{e, (123), (132)\}$. Since $|S_3| = 6$ and $|H| = 3$,

$$[S_3 : H] = 6/3 = 2.$$

The two cosets are H and $(23)H = \{(23), (12), (13)\}$. They separate the rotations from the reflections of the triangle.

Group Theory

Element order divides group order

For any $g \in G$, $\langle g \rangle$ is a subgroup and $|\langle g \rangle| = \text{ord}(g)$. Lagrange gives

$$\boxed{\text{ord}(g) \mid |G|}.$$

For S_3 , possible element orders must divide 6. The actual possibilities are 1, 2, and 3. The theorem gives a necessary condition, without asserting that every divisor occurs as an element order.

Group Theory

A useful consequence

Because $\text{ord}(g)$ divides $|G|$ and $g^{\text{ord}(g)} = e$,

$$\boxed{g^{|G|} = e} \quad \text{for every } g \in G.$$

For example, every element of a group of order 6 satisfies $g^6 = e$. This does not say that every element has order 6.

Group Theory

Prime order groups

Let $|G| = p$ be prime. For $g \neq e$, Lagrange says $\text{ord}(g)$ divides p . Since $\text{ord}(g) > 1$, necessarily $\text{ord}(g) = p$. The subgroup $\langle g \rangle$ therefore contains all p elements:

$$G = \langle g \rangle \cong Z_p.$$

Every group of prime order is cyclic. Every nonidentity element generates it.

Group Theory

Cosets of a reflection subgroup

Let $G = D_3$ and $H = \{e, s\}$, where $r^3 = s^2 = e$ and $sr s = r^{-1}$. The left cosets are

$$H = \{e, s\}, \quad rH = \{r, rs\}, \quad r^2H = \{r^2, r^2s\}.$$

They partition all six elements, so $[D_3 : H] = 3$.

Group Theory

Left and right need not agree

Right cosets have the form $Hg = \{hg : h \in H\}$. For the same subgroup $H = \{e, s\}$,

$$rH = \{r, rs\}, \quad Hr = \{r, sr\} = \{r, r^2s\}.$$

These are different. The distinction matters when we study *normal subgroups* and quotient groups in Week 3.

Group Theory

When do two elements label the same coset?

$$aH = bH \iff b^{-1}a \in H.$$

If $aH = bH$, then $a = ae \in bH$. Hence $a = bh$ for some $h \in H$, giving $b^{-1}a = h \in H$.
Conversely, if $b^{-1}a = h \in H$, then

$$aH = bhH = bH.$$

In Z_6 with $H = \{0, 3\}$, we have $1 + H = 4 + H$ because $1 - 4 = -3 = 3 \pmod{6}$ belongs to H .

Group Theory

Check your understanding

In Z_8 , let $H = \langle 2 \rangle = \{0, 2, 4, 6\}$.

1. List all distinct cosets of H .
2. Calculate $[Z_8 : H]$.
3. What are the possible orders of elements of Z_8 according to Lagrange? Which occur?

Group Theory

Check your understanding: solution

$$0 + H = \{0, 2, 4, 6\}, \quad 1 + H = \{1, 3, 5, 7\}.$$

There are two cosets, so $[Z_8 : H] = 2$. Lagrange allows orders 1, 2, 4, 8. All occur: 0 has order 1, 4 has order 2, 2 has order 4, and 1 has order 8.

Group Theory

Exercise: what does Lagrange tell us?

1. A finite group G has 30 elements. Could it contain a subgroup of order 8?
2. The divisors of $|S_3| = 6$ are 1, 2, 3, 6. Which occur as element orders?
3. Does S_3 contain an element that generates the whole group?

Explain which conclusions follow from Lagrange and which require examining the particular group.

Group Theory

Solution: necessary conditions

An order-8 subgroup of a group of order 30 is impossible because $8 \nmid 30$. In S_3 , the identity has order 1, transpositions have order 2, and three-cycles have order 3. There is no element of order 6, so S_3 is not cyclic. Although $6 \mid |S_3|$, this divisibility does not produce an element of order 6.

Group Theory

Direct products of groups

The **direct product** $G \times K$ consists of pairs (g, k) with $g \in G$ and $k \in K$. Multiplication is componentwise:

$$(g_1, k_1)(g_2, k_2) = (g_1g_2, k_1k_2).$$

Closure and associativity hold in each component. The identity and inverses are

$$e_{G \times K} = (e_G, e_K), \quad (g, k)^{-1} = (g^{-1}, k^{-1}).$$

For finite groups, $|G \times K| = |G| |K|$.

Textbook: Zee, I.1, pp. 44–45; Zee uses $\otimes$ for this group product.

Group Theory

$Z_2 \times Z_3$ is cyclic

Define $\psi : Z_6 \rightarrow Z_2 \times Z_3$ by

$$\psi([k]_6) = ([k]_2, [k]_3).$$

It is well defined because 6 is divisible by both 2 and 3, and it preserves addition componentwise. If $\psi([k]_6) = (0, 0)$, then k is divisible by both 2 and 3, hence by 6. Therefore $\ker \psi = \{0\}$, so ψ is injective. Both groups have six elements, so ψ is also surjective:

$$\boxed{Z_6 \cong Z_2 \times Z_3.}$$

The element $(1, 1) = \psi([1]_6)$ generates the product group.

Group Theory

$Z_2 \times Z_2$ and the group D_2

The product has four elements: $(0, 0)$, $(1, 0)$, $(0, 1)$, $(1, 1)$. Bünemann's D_2 consists of the identity and the half-turns about three perpendicular axes:

$$D_2 = \{e, u, v, uv\}, \quad u^2 = v^2 = e, \quad uv = vu.$$

Define $\theta : Z_2 \times Z_2 \rightarrow D_2$ by $\theta(a, b) = u^a v^b$. The relations make this well defined and multiplication preserving. It maps the four pairs to four distinct elements, so

$$\boxed{Z_2 \times Z_2 \cong D_2.}$$

Textbook: Zee calls this abstract group V : I.1, p. 45. Compare Bünemann, §2.2.1.

Group Theory

Why $Z_2 \times Z_2$ is not Z_4

Every nonidentity element of $Z_2 \times Z_2$ has order 2:

$$(a, b) + (a, b) = (0, 0).$$

But Z_4 contains elements of order 4, such as $[1]_4$. An isomorphism would preserve element orders. Therefore

$$Z_2 \times Z_2 \not\cong Z_4.$$

Having the same number of elements does not make two groups isomorphic. Their multiplication structure matters.

Group Theory

The first two weeks

- Symmetries form groups; Z_n , D_n and S_3 give concrete examples.
- Subgroups inherit the operation. Homomorphisms preserve it; kernels detect injectivity.
- Cayley's theorem realizes finite groups as permutation groups.
- Cosets partition a group and prove Lagrange's theorem.
- Direct products combine groups: $Z_2 \times Z_3 \cong Z_6$, but $Z_2 \times Z_2 \not\cong Z_4$.

Next: normal subgroups, quotient groups, conjugacy classes and further permutation groups.

Reading: Zee, I.1, pp. 43–45; Bünemann, §2.3.4, pp. 21–22.